

Gas Turbine Emission & Energy Analysis

Data Analytics Project Work

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1. Dataset Description & Background

- **Context:** The dataset comprises 36,733 hourly aggregated records of 11 sensor measurements.
- **Industrial Background:** Data was collected from an operational gas turbine located in the north-western region of Turkey across five consecutive years (2011 to 2015).
- **Primary Objectives:**
 - 1 **Energy-yield prediction:** Predict Turbine Energy Yield (TEY) from operating and ambient measurements.
 - 2 **Emissions analysis:** Study and model Carbon Monoxide (CO) and Nitrogen Oxides (NOX).

1. Variables & Engineering Roles

- **Ambient Predictors:**

- **AT ($^{\circ}\text{C}$):** Ambient intake air temperature; influences air density.
- **AP (mbar) & AH (%):** Atmospheric pressure and relative humidity.

- **Turbine Predictors:**

- **AFDP, GTEP, CDP (mbar):** Pressure drops and compressor discharge pressures.
- **TIT & TAT ($^{\circ}\text{C}$):** Turbine inlet (firing) and after (exhaust) temperatures.

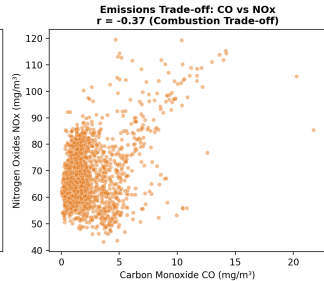
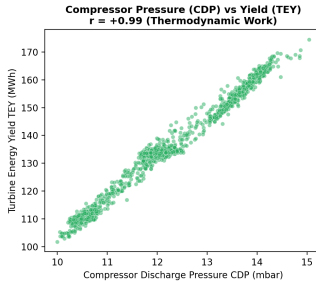
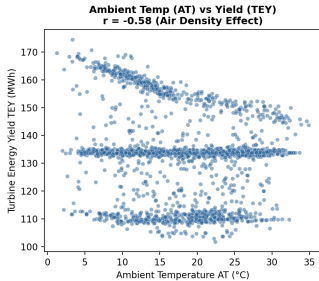
- **Target Variables:**

- **TEY (MWh):** Hourly net electrical energy output.
- **CO & NOX (mg/m^3):** Measured flue gas concentrations.

2. Data Cleaning & Partitioning

- **Data Quality:** 0 missing values detected. 7 duplicate sensor profiles were removed, leaving 36,726 clean observations.
- **Outliers:** Evaluated via robust rank-based dispersion (IQR / 1.35). Anomalies (e.g., 5.29% in TAT) were kept to preserve real transient operating conditions.
- **Chronological Partitioning:**
 - **Train/CV Set:** 2011–2013 (22,191 observations, 60.42%).
 - **Test Set:** 2014–2015 (14,535 observations, 39.58%).
- **Standardization:** Z-score normalization was fitted strictly on the training set to prevent data leakage.

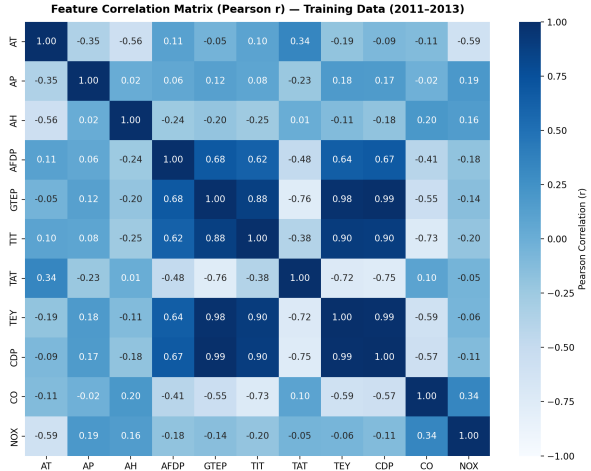
3. Exploratory Analysis: Physical Relationships



- **Air Density Effect:** Negative correlation between Ambient Temp and Yield.
- **Thermodynamic Work:** Near-perfect linear correlation between Compressor Pressure and Yield.
- **Emissions Relationship:** Positive correlation observed between CO and NOx ($r = 0.34$).

3. Exploratory Analysis: Pearson Correlation

- The Pearson correlation matrix evaluates linear dependence.
- **High Collinearity:** CDP, GTEP, and TIT are highly correlated with each other and the target TEY.
- TAT exhibits strong negative correlations with pressure variables.
- CO and NOX show a positive correlation, confirming linked combustion dynamics.



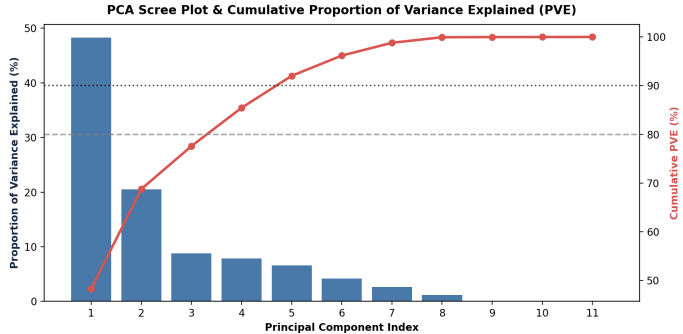
3. Exploratory Analysis: PCA Variance

- **Dimensionality Reduction:**

PCA was applied to handle multicollinearity.

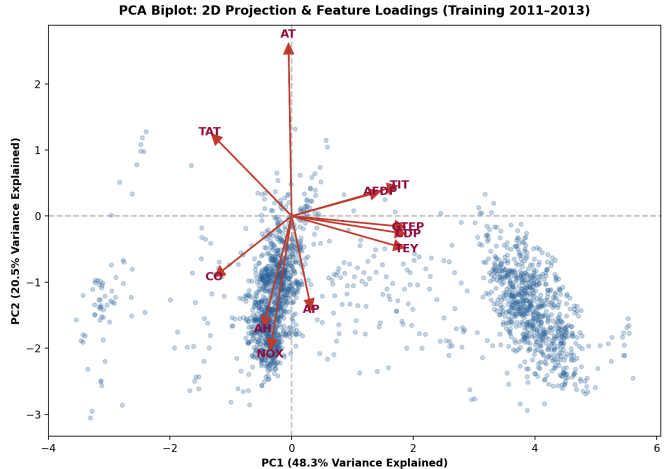
- **Scree Plot:** Visualizes the Proportion of Variance Explained (PVE).

- PC1 alone explains 48.28% of the variance, while PC1 and PC2 together capture 68.76% of the total variance.



3. Exploratory Analysis: PCA 2D Projection

- **Biplot Analysis:** Shows 2D projection and feature loadings.
- **Feature Groupings:** Thermodynamic features (TEY, CDP, GTEP, TIT) strongly pull together along the PC1 axis.
- Ambient variables and emissions project distinctly, showing different underlying variances.



4. Main Analysis: Multicollinearity Diagnostics

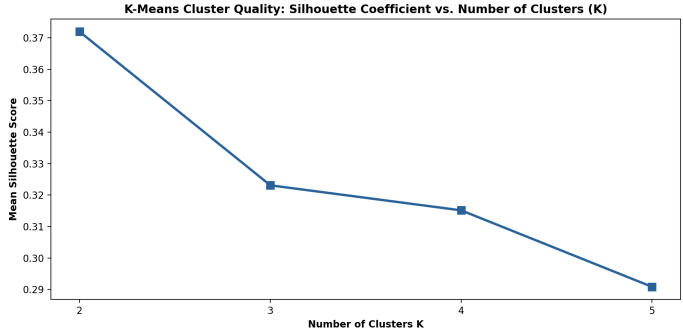
- **Variance Inflation Factor (VIF):** Used to diagnose multicollinearity. A VIF greater than 10 indicates severe multicollinearity.

Predictor Feature	VIF Score	Collinearity Assessment
Ambient Temperature (AT)	3.04	Low / Acceptable
Ambient Pressure (AP)	1.47	Low / Acceptable
Ambient Humidity (AH)	1.71	Low / Acceptable
Air Filter Diff. Pressure (AFDP)	1.95	Low / Acceptable
Exhaust Pressure (GTEP)	260.47	Severe (VIF > 10)
Inlet Temp (TIT)	262.12	Severe (VIF > 10)
After Temp (TAT)	133.21	Severe (VIF > 10)
Discharge Pressure (CDP)	342.82	Severe (VIF > 10)

- **Solution:** Principal Component Regression (PCR) was adopted to overcome coefficient variance inflation.

4. Main Analysis: K-Means Clustering

- **Objective:** Unsupervised regime discovery.
- **Quality Metric:** Mean Silhouette Coefficient vs. Number of Clusters (K).
- Although $K = 2$ had a higher score (0.3763), $K = 3$ (0.3023) was retained to obtain three physically interpretable operating regimes.



4. Main Analysis: Operational Regimes

- **Cluster Profiles:** The K-Means algorithm ($K = 3$) partitioned the operational space into distinct dispatch states.

Regime Name	TEY	TIT	CDP	AT	CO	NOX	Hours
Peak Load	158.16	1099.61	13.8	13.14	1.00	67.72	4,309
Nominal Load	134.45	1090.73	12.2	24.50	1.48	65.04	7,820
Part Load	122.28	1070.06	11.2	14.39	3.30	72.13	10,062

- **Insight:** Part Load operations correlate with higher CO and NOX emissions simultaneously, confirming the positive relationship between these pollutants at lower operational states.

5-6. Predictive Model Summary

Target	Model	Train R^2	Train RMSE	Test R^2	Test RMSE
TEY	OLS	0.998	0.76	0.950	3.35
TEY	PCR (4 PCs)	0.969	2.81	0.940	3.68
CO	Polynomial deg 2	0.648	1.36	0.264	1.88
NOX	OLS	0.454	8.16	-1.093	15.30
TEY	Baseline	0.000	16.03	-0.000	14.98
CO	Baseline	0.000	2.30	-0.033	2.23
NOX	Baseline	0.000	11.04	-0.691	13.76

- **Yield:** TEY is predicted accurately by both OLS and PCR.
- **Emissions:** OLS fails to generalize for NOX (negative Test R^2), and CO polynomial struggles on test data, proving standard linear models are insufficient.

7. Conclusions

- **Methodological Consistency:** Chronological splitting prevented data leakage, and PCR effectively handled the severe multicollinearity ($VIF > 260$) among turbine predictors.
- **Energy Yield (TEY):** Readily predictable due to robust thermodynamic relationships.
- **Emissions Modeling:** CO and NOX models suffered from complex, non-linear emission dynamics. Standard linear methods are inadequate.
- **Next Steps:** Non-linear algorithms (Random Forest, XGBoost) and regime-specific models (leveraging our $K=3$ clusters) are required to capture emission dynamics accurately.

Thank You

Questions?